

APPASA Strategy Document

Stage 2 – Prepare: Data Understanding & Source Assessment



APPASA: Prepare Stage

Introduction

I will use this **APPASA strategy document** to record my decisions and reflections as I work through **Stage 2: Prepare** of this project. This document will serve as a guide, helping me consider my responses and reflections throughout this stage of the data analysis process.

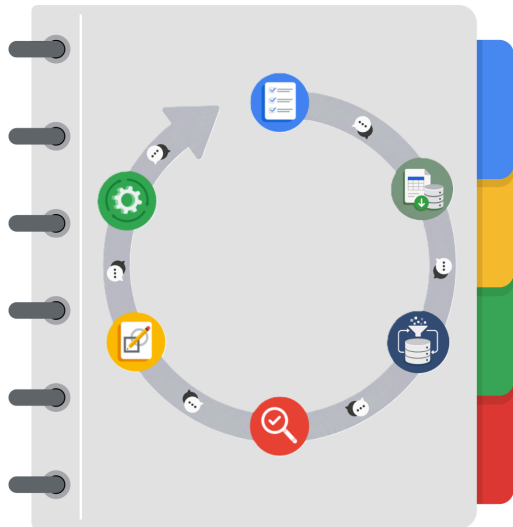
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. These questions are primarily situated within the Prepare stage of the APPASA workflow, focusing on understanding the dataset, evaluating its credibility, assessing its suitability for analysis, and identifying potential data quality considerations before processing begins. I will address each question by drawing on relevant materials from the project notebook, SQL scripts, the APPASA framework, and best practices for data understanding and source assessment.





Data Project Questions & Considerations

Data Source

- What dataset(s) will be used for this project?

This project uses **Cyclistic's historical bike trip data**, represented by the **twelve monthly Divvy trip datasets** covering **January 2024 through December 2024**. Each monthly dataset is distributed as a compressed ZIP archive containing a CSV file of individual bike trip records. Together, these twelve monthly datasets represent a complete year of bike-share activity and collectively form the primary dataset used throughout this analysis.

Each trip record represents a single bike trip and contains information including the ride identifier, bicycle type, trip start and end timestamps, station names and identifiers, geographic coordinates, and rider classification (annual member or casual rider). Collectively, these variables provide the information required to examine rider behavior, identify usage patterns, compare annual members with casual riders, and answer the project's primary business question.

- Where is the data located, and how will it be stored?

The dataset is publicly available through the **official Divvy Trip Data repository**, where historical Divvy bike trip data is distributed as monthly ZIP archives containing CSV files. For this project, I downloaded the twelve monthly datasets covering **January 2024 through December 2024** directly from the official repository.

After downloading, the datasets will be organized within a structured local project directory, preserving both the original ZIP archives and the extracted CSV files. The extracted CSV files will then be imported into a **MySQL** database for data ingestion, preprocessing, validation, feature engineering, and analytical querying. Throughout the project, SQL scripts, project documentation, Jupyter notebooks, and supporting assets will be maintained within the project repository, while the processed data will remain stored in the MySQL database to support a reproducible, organized, and efficient analytical workflow.

Data Governance

- Who published the dataset, and what is its source?

The dataset used in this project is publicly available through the **official Divvy Trip Data repository**, which serves as the official source for historical Divvy bike trip data. The files are hosted on an Amazon S3 storage bucket and distributed as monthly ZIP archives containing trip-level CSV datasets.

The **Google Data Analytics Capstone** states that the dataset is made available by **Motivate International Inc.** under the applicable usage license. According to the current **Divvy Data License Agreement**, however,

the **City of Chicago** owns the dataset, while **Lyft Bikes and Scooters, LLC.**, the current operator of the Divvy bike-share system, makes the data publicly available through the official Divvy data portal.

For this project, I used the twelve monthly datasets covering **January 2024 through December 2024**, which provide comprehensive records of bike-share trips over a one-year period.

The Google Data Analytics Capstone references these publicly available Divvy datasets because the fictional company **Cyclistic** is modeled using real-world Divvy bike-share data. As a result, the Divvy trip datasets provide an appropriate and representative data source for answering the project's business questions.

- How are licensing, privacy, security, and accessibility addressed for this dataset?

The dataset used in this project is publicly available through the official Divvy Trip Data repository and is distributed under the **Divvy Data License Agreement**. The license grants users a non-exclusive, royalty-free, perpetual license to access, reproduce, analyze, modify, and use the data for lawful purposes, including educational and analytical projects. However, the license prohibits redistributing the dataset as a standalone resource, attempting to identify individual riders, circumventing data access restrictions, or implying any affiliation with or endorsement by Divvy, Lyft Bikes and Scooters, LLC, or the City of Chicago.

The dataset has been anonymized and does not contain any personally identifiable information (PII). In accordance with the Divvy Data License Agreement, users must not attempt to correlate the trip data with names, addresses, payment information, or any other information that could identify individual riders. This ensures that rider privacy is protected while still enabling meaningful behavioral analysis.

To maintain data security throughout this project, the downloaded datasets will be stored within a structured local project directory and imported into a MySQL database for analysis. The original source files will be preserved separately from the processed data to maintain data integrity, support reproducibility, and provide an auditable record of the raw data. Throughout the project, only aggregated analytical results, visualizations, and business insights will be presented, ensuring that the analysis remains ethical, privacy-conscious, and compliant with the dataset's licensing terms.

Because the dataset is publicly available through the official Divvy Trip Data repository, it is readily accessible for educational, research, and analytical purposes. This accessibility enables the dataset to be downloaded, reproduced, and analyzed while supporting transparent, reproducible, and well-documented analytics workflows.

Dataset Understanding

- Why is this dataset appropriate for answering the business question?

This dataset is appropriate for answering the project's business question because it contains detailed historical records of individual bike trips made by both **annual members** and **casual riders**, the two customer groups central to this analysis. Each trip record includes attributes such as rider classification, bicycle type,

trip start and end timestamps, station information, and geographic coordinates, providing the information necessary to compare how the two rider groups use Cyclistic bikes.

By covering a complete twelve-month period from **January 2024 through December 2024**, the dataset captures seasonal variations, weekday and weekend travel patterns, ride durations, station usage, and other behavioral trends. This comprehensive temporal coverage enables meaningful comparisons between annual members and casual riders while minimizing the influence of short-term fluctuations or seasonal bias.

The dataset represents real-world operational bike-share data collected from the Divvy system, which serves as the basis for the fictional Cyclistic case study in the Google Data Analytics Capstone. As a result, it provides a realistic and representative foundation for identifying rider behavior patterns, generating actionable business insights, and developing data-driven marketing strategies aimed at converting casual riders into annual members.

- How is the dataset organized?

The dataset is organized as **twelve separate monthly CSV files**, each representing all recorded Divvy bike trips for a specific month between **January 2024 and December 2024**. Together, these monthly datasets form a complete annual dataset that captures one full year of bike-share activity.

Each CSV file follows a **consistent tabular structure**, where **each row represents a single bike trip** and **each column represents a specific attribute** describing that trip. The dataset contains 13 variables, including the ride identifier, bicycle type, trip start and end timestamps, station names and identifiers, geographic coordinates, and rider classification (annual member or casual rider).

Because every monthly dataset shares the same schema, the files can be seamlessly combined into a single consolidated annual dataset without requiring structural modifications. This consistent organization supports efficient data ingestion, preprocessing, validation, feature engineering, analytical querying, and visualization throughout the project's analytical workflow.

- What files comprise the dataset?

The dataset comprises **twelve monthly Divvy trip datasets**, each distributed as a compressed ZIP archive containing a CSV file of individual bike trip records. Collectively, these files represent a complete year of historical bike-share activity spanning **January 2024 through December 2024**.

The dataset includes the following monthly CSV files:

- 202401-divvy-tripdata.csv
- 202402-divvy-tripdata.csv
- 202403-divvy-tripdata.csv



- 202404-divvy-tripdata.csv
- 202405-divvy-tripdata.csv
- 202406-divvy-tripdata.csv
- 202407-divvy-tripdata.csv
- 202408-divvy-tripdata.csv
- 202409-divvy-tripdata.csv
- 202410-divvy-tripdata.csv
- 202411-divvy-tripdata.csv
- 202412-divvy-tripdata.csv

Each file contains all recorded bike trips for its respective month and follows an identical schema consisting of **13 variables**. This consistent structure enables the monthly datasets to be efficiently combined into a single annual dataset for preprocessing, analysis, visualization, and reporting while maintaining schema consistency and data integrity throughout the analytical workflow.

- What variables are available for analysis?

The dataset contains **13 variables** that collectively describe the characteristics of each recorded bike trip. These variables provide the information necessary to analyze rider behavior, usage patterns, and operational characteristics of the Cylcistic bike-share system.

The available variables can be grouped into the following categories:

- **Trip Identification**

- **ride_id** – Unique identifier assigned to each bike trip.

- **Bicycle Information**

- **rideable_type** – Type of bicycle used during the trip.

- **Temporal Information**

- **started_at** – Date and time when the trip began.

- **ended_at** – Date and time when the trip ended.

- **Station Information**

- **start_station_name** – Name of the origin station.
- **start_station_id** – Unique identifier of the origin station.
- **end_station_name** – Name of the destination station.
- **end_station_id** – Unique identifier of the destination station.

- **Geographic Information**

- **start_lat** – Latitude of the trip origin.
- **start_lng** – Longitude of the trip origin.
- **end_lat** – Latitude of the trip destination.
- **end_lng** – Longitude of the trip destination.

- **Rider Classification**

- **member_casual** – Indicates whether the rider is an annual member or a casual rider.

Collectively, these variables provide the information required to examine temporal, spatial, and behavioral patterns, compare annual members with casual riders, and answer the project's primary business question.

- What is the data dictionary for this project?

The data dictionary is a structured reference that documents every variable contained in the dataset, including its name, data type, description, and purpose. It serves as a central source of metadata, ensuring a consistent understanding of the dataset throughout the project's lifecycle while supporting accurate interpretation, preprocessing, analysis, and reporting.

For this project, the data dictionary documents the 13 variables contained within the Divvy trip dataset, including ride identifiers, bicycle types, trip timestamps, station information, geographic coordinates, and rider classifications. It provides a standardized reference for understanding each variable's data type, meaning, and intended use, helping maintain consistency across SQL queries, feature engineering, visualizations, and business reporting.

The table below presents the complete data dictionary for the variables used throughout this project.



Variable	Data Type	Description
ride_id	VARCHAR	Unique identifier assigned to each bike trip.
rideable_type	VARCHAR	Type of bicycle used for the trip (e.g., classic bike or electric bike).
started_at	TIMESTAMP	Date and time when the trip began.
ended_at	TIMESTAMP	Date and time when the trip ended.
start_station_name	VARCHAR	Name of the station where the trip originated.
start_station_id	VARCHAR	Unique identifier of the origin station.
end_station_name	VARCHAR	Name of the station where the trip ended.
end_station_id	VARCHAR	Unique identifier of the destination station.
start_lat	DOUBLE	Latitude coordinate of the trip origin.
start_lng	DOUBLE	Longitude coordinate of the trip origin.
end_lat	DOUBLE	Latitude coordinate of the trip destination.
end_lng	DOUBLE	Longitude coordinate of the trip destination.
member_casual	VARCHAR	Rider classification indicating whether the rider is an annual member or a casual rider.



Data Credibility

- Does the dataset satisfy the ROCCC framework?

The **ROCCC framework** is used to evaluate the credibility and suitability of a dataset by assessing whether it is **Reliable, Original, Comprehensive, Current, and Cited**. Based on this evaluation, the Divvy trip dataset satisfies all five components of the ROCCC framework and provides a credible foundation for answering the project's business question.

Reliable:

The dataset consists of real-world operational records generated by the Divvy bike-share system and is publicly distributed through the official Divvy Trip Data repository. Its official origin, documented licensing, and widespread use in the Google Data Analytics Capstone make it a reliable source for behavioral analysis.

Original:

The data is obtained directly from the official Divvy Trip Data repository rather than a third-party aggregator or secondary source. This ensures that the dataset represents the original operational records collected by the bike-share system.

Comprehensive:

The dataset contains detailed trip-level information, including ride identifiers, bicycle types, trip timestamps, station information, geographic coordinates, and rider classifications. By covering a complete twelve-month period from **January 2024 through December 2024**, it provides sufficient temporal coverage to analyze rider behavior across different seasons and usage patterns.

Current:

The project uses the most recent complete calendar year available at the time the analysis was conducted, covering **January 2024 through December 2024**. This provides a recent and representative view of rider behavior while capturing seasonal variations across an entire year.

Cited:

The dataset is publicly available through the official Divvy Trip Data repository and is referenced by the Google Data Analytics Capstone as the data source for the Cyclistic case study. Its source, ownership, publisher, and licensing information are clearly documented, allowing the dataset to be properly cited and independently verified.

Based on this evaluation, the dataset satisfies all five components of the ROCCC framework and is well suited for conducting reliable, reproducible, and data-driven analysis.

- Are there any potential sources of bias or limitations within the dataset?

Although the dataset is comprehensive and well suited for this analysis, several limitations should be considered when interpreting the results.

The dataset has been **anonymized** and does not contain personally identifiable information (PII) or demographic attributes such as age, gender, income, occupation, or place of residence. As a result, the analysis can identify behavioral differences between annual members and casual riders but cannot determine how demographic or socioeconomic factors influence those differences.

The dataset records **observed bike trips** rather than customer motivations or preferences. Consequently, it can reveal **how** annual members and casual riders behave differently but cannot directly explain **why** those behavioral differences exist or what factors motivate casual riders to purchase an annual membership.

Additionally, the analysis is based on historical trip records covering a single twelve-month period from **January 2024 through December 2024**. While this provides complete seasonal coverage for one calendar year, changes in rider behavior, pricing policies, infrastructure, or external conditions outside this period may influence future travel patterns.

Finally, the dataset describes completed bike trips and their associated characteristics. It does not include complementary business information such as marketing campaign exposure, customer feedback, membership purchase history, or financial metrics, limiting the ability to directly evaluate the effectiveness of specific marketing strategies or the profitability of individual rider segments.

These limitations do not prevent meaningful analysis but should be considered when interpreting the findings and developing business recommendations.

- Are there any ethical considerations associated with using this dataset?

Yes. Although the dataset is publicly available and anonymized, several ethical considerations should be observed throughout the project.

The dataset does not contain personally identifiable information (PII), and rider privacy must be respected at all times. In accordance with the **Divvy Data License Agreement**, no attempt should be made to correlate the trip records with names, addresses, payment information, or any other external data that could be used to identify individual riders.

The analysis should focus on **aggregated trends and behavioral patterns** rather than individual trips or riders. All findings, visualizations, and recommendations should be presented at an aggregate level to ensure that no individual can be identified, directly or indirectly.

It is also important to interpret the results objectively and avoid drawing conclusions that are not supported by the available data. Since the dataset records only observed trip behavior, the analysis should not make unsupported assumptions about riders' motivations, preferences, demographics, or personal circumstances.

Finally, the dataset should be used in accordance with the **Divvy Data License Agreement**, respecting its licensing conditions and usage restrictions while ensuring that all analyses, reports, and project deliverables are conducted responsibly and ethically.



Data Suitability & Readiness

- How does this dataset support the project's business objective?

The primary business objective of this project is to support Cyclistic's goal of increasing the number of annual memberships by understanding how annual members and casual riders use the bike-share system differently. The dataset directly supports this objective by providing detailed historical records of bike trips taken by both rider groups over a complete twelve-month period.

Because each trip record includes rider classification, bicycle type, trip timing, station information, and geographic coordinates, the dataset enables meaningful comparisons of riding behavior, travel patterns, trip durations, bicycle preferences, and station usage between annual members and casual riders. These behavioral insights can help identify opportunities to encourage casual riders to transition to annual memberships.

By transforming raw trip data into actionable business insights, the dataset provides the evidence needed to develop targeted marketing strategies, support data-driven decision-making, and formulate recommendations that align with Cyclistic's long-term objective of increasing annual membership adoption.

- How was the data's integrity initially verified?

Before beginning data processing, the dataset's integrity was initially verified by confirming that all required files were successfully downloaded from the official Divvy Trip Data repository and corresponded to the intended analysis period of **January 2024 through December 2024**. Each monthly dataset was extracted from its ZIP archive and inspected to ensure that the files were complete, accessible, and free from corruption.

An initial review was then conducted to verify that all twelve CSV files shared a consistent schema, including identical variable names, data types, and column ordering. The datasets were also checked to confirm that each file contained trip-level records with the expected structure and could be successfully imported into the project's analytical environment without structural issues.

These initial verification steps established that the dataset was complete, structurally consistent, and ready to proceed to the data processing stage, where detailed data quality assessments and cleaning operations would be performed.

- What potential data quality issues should be expected before processing begins?

Before beginning the data processing stage, several potential data quality issues can reasonably be anticipated due to the size and operational nature of the dataset. These expectations help guide the data validation and cleaning process that follows.

Potential issues include:



- **Missing values:** Certain fields, such as station names, station identifiers, or geographic coordinates, may contain missing or incomplete information.
- **Duplicate records:** Duplicate trip records may exist because of data collection or processing anomalies and should be identified during validation.
- **Invalid or inconsistent timestamps:** Trip start and end timestamps should be examined for formatting inconsistencies or unrealistic ride durations.
- **Data type inconsistencies:** Variables should be verified to ensure they conform to their expected data types and formats.
- **Schema consistency:** Although the monthly datasets are expected to share an identical schema, the column structure and naming conventions should be verified before combining the files.
- **Incomplete or erroneous records:** Some trip records may contain incomplete, inconsistent, or invalid values that could affect the accuracy of subsequent analyses.

These potential issues represent expected data quality considerations rather than confirmed findings. A comprehensive assessment of data quality will be performed during the **Process** stage before any analytical activities begin.

- What assumptions do I currently have about the dataset before beginning the processing stage?

Before beginning the data processing stage, I assume that the dataset is generally well structured, complete, and suitable for analysis because it is obtained directly from the official Divvy Trip Data repository and is used as the primary dataset for the Google Data Analytics Capstone.

I also assume that the twelve monthly datasets share a consistent schema, allowing them to be combined into a single annual dataset without requiring significant structural modifications. Although I expect the dataset to be of high quality, I anticipate encountering common real-world data quality issues such as missing values, duplicate records, inconsistent timestamps, or incomplete trip information that will require validation and cleaning during the Process stage.

Finally, I assume that the available variables contain sufficient information to identify meaningful behavioral differences between annual members and casual riders. However, I recognize that these assumptions must be verified through systematic data exploration, validation, and preprocessing before any analysis or business conclusions are drawn.